

Indiegogo launches 'generosity'

The charity crowdfunding platform of Indiegogo has rebranded to 'Generosity' and is a standalone site now. <http://dgit.in/Nv15aw>



Instagram's ad increased!

Instagram has seen a huge influx of advertisements and 50% of its biggest fans seem to be unhappy with it. <http://dgit.in/Nv15ax>

Industry Connect



3D printing in medical applications

We chat with Srinivas Shastri of Altem Technologies about the recent trends in 3D printing and the upcoming materials used in the process.



◀ Srinivas Shastri, Technical Director, Altem Technologies Pvt. Ltd.

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d We have seen that 3D printed materials don't have a smooth polished finish, so, how is it improving?

Srinivas: 3D printing has evolved a lot in the last decade. I completely agree with you, when initially the parts were printed maybe 10-12 years ago, the finish was not so great or it was just one material that was used, the consumer didn't have much of a choice. Some of them were really concerned about that aspect but with the new materials and technologies getting better, it's improving. It all depends on what technology you use and for what application. Some application may not need a completely smooth surface but only requires it to be functional for a mechanical application on a shop floor or on a vehicle where the external surface doesn't really affect performance. The printing technology has really evolved over the years. Current printers don't just use one material, you can now have multiple materials which are transparent, flexible and colourful. There is an acrylic based resin which is photo curable, so this resin is jetted out and exposed to UV light into a hardened state. Definitely technology has evolved a lot in the last ten years and now you can get almost 1,800 combinations using only 14 basic materials.

d Can complex assemblies with moving parts be 3D printed?

Srinivas: Moving parts are all done in a single print. 3D printers are capable of printing parts with a moving assembly. 3D printers come with an option of having a support material, which provides support to any overhangs or any voids which are there in the system. So, basically the void is filled with a temporary material called a support material. The 3D printer software decides the location of the voids, then the software fills it up with the temporary material. Eventually when the part is built, it is dropped into water or washed away with a jet.

d Other than industrial products for what other applications can 3D printers be used?

Srinivas: All these years, it was mainly in the engineering domain that 3D printer was being used, but in the last couple of years there has been a lot of attraction in medical and dental applications. There are a lot of 3D printed organs printed as pre-surgical models. So, if someone has to be operated upon, they are scanned and the data obtained is converted into a 3D model, then printed to analyse the problem. So, the doctors can use the same model which is a replica of the patient's organ and then they will be able to diagnose the problem better since they

now have access to data which they are definitely gonna miss using MRI or CT because it's always a 2D view. Using an actual 3D printed part reduces the time spent on planning and analysing the operating. Another application is in the dental industry. There can be implants or ridges which are 3D printed metal using your prosthodontics and orthodontics. All the corrections can be modeled with a 3D printer and the entire workflow is digitised. That's where we see a major growth, in fact, Stratasys has a range of dental portfolio printers just addressing the dental market. We are moving away from steel braces to a clear 3D printed material so, you don't even notice the braces. I think this is where the future lies other than engineering applications, automotive and aerospace where it is already being conducted. The upcoming 3D printers are more likely to have life changing applications in medical applications.

d So, in a few years will we have 3D printed organ transplants?

Srinivas: That's the future everybody is thinking about but then I am thinking we are far away from that because we would need materials which are very biocompatible and doesn't really damage your other organs. For now, I don't think we're close to achieving this but maybe in the future we will accomplish 3D printed organs. **d**